Q1

\# of poles : 6 $\Rightarrow$ pole pair $p = 3$

\# of slots : 72 (Q) 3 phase, double layer

Full Pitched :

slot angle $\alpha = \dfrac{2\pi}{(Q/p)} = 15°$

$q = \dfrac{Q}{2 \cdot m \cdot p}$, m = # of phase $\Rightarrow q = 4$

Distribution factor $k_d = \dfrac{\sin\left(q \cdot \frac{\alpha}{2}\right)}{q \cdot \sin\left(\frac{\alpha}{2}\right)} = 0{,}957$

Pitch factor $k_p = \sin\left(\frac{\lambda}{2}\right)$, λ: coil pitch

$k_p = \sin\left(\frac{\pi}{2}\right) = 1$

Winding factor $k_w = k_p \cdot k_d = 0{,}957$

for 3rd harmonics :

$k_d = \dfrac{\sin\left(n q \frac{\alpha}{2}\right)}{q \cdot \sin\left(n \frac{\alpha}{2}\right)} = 0{,}653$ $k_p = \sin\left(\frac{3\pi}{2}\right) = -1 \Rightarrow k_w = -0{,}653$

for 5th harmonics :

$k_d = 0{,}205$ $k_p = 1$ $\Rightarrow k_w = 0{,}205$

Winding Diagram : Slots per pole pairs : $72/3 = 24$

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	A_1^+	A_2^+	A_3^+	A_4^+	C_1^-	C_2^-	C_3^-	C_4^-	B_1^+	B_2^+	B_3^+	B_4^+	A_1^-	A_2^-	A_3^-	A_4^-	C_1^+	C_2^+	C_3^+	C_4^+	B_1^-	B_2^-	B_3^-	B_4^-
	A_1^+	A_2^+	A_3^+	A_4^+	C_1^-	C_2^-	C_3^-	C_4^-	B_1^+	B_2^+	B_3^+	B_4^+	A_1^-	A_2^-	A_3^-	A_4^-	C_1^+	C_2^+	C_3^+	C_4^+	B_1^-	B_2^-	B_3^-	B_4^-

11/12 short - pitched :

k_d is same above, $k_d = \dfrac{\sin\left(nq\frac{\alpha}{2}\right)}{q\cdot\sin\left(n\frac{\alpha}{2}\right)}$

k_p: $\lambda = 165^\circ \Rightarrow \quad k_p = \sin\left(\frac{\lambda}{2}\right) = 0{,}991$

$k_w = k_p \cdot k_d = 0{,}991 \cdot 0{,}957 = 0{,}948$

for 3rd harmonics:

$k_d = 0{,}653$, $k_p = 0{,}991 \Rightarrow k_w = 0{,}647$

for 5th harmonics:

$k_d = 0{,}205 \quad k_p = 0{,}991 \Rightarrow k_w = 0{,}203$

Winding Diagram :

Q	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24
	A_1^+	A_2^+	A_3^+	A_4^+	C_1^-	C_2^-	C_3^-	C_4^-	B_1^+	B_2^+	B_3^+	B_4^+	A_5^-	A_6^-	A_7^-	A_8^-	C_5^+	C_6^+	C_7^+	C_8^+	B_5^-	B_6^-	B_7^-	B_8^-
	A_6^+	A_7^+	A_8^+	C_5^-	C_6^-	C_7^-	C_8^-	B_5^+	B_6^+	B_7^+	B_8^+	A_1^-	A_2^-	A_3^-	A_4^-	C_1^+	C_2^+	C_3^+	C_4^+	B_1^-	B_2^-	B_3^-	B_4^-	A_5^+

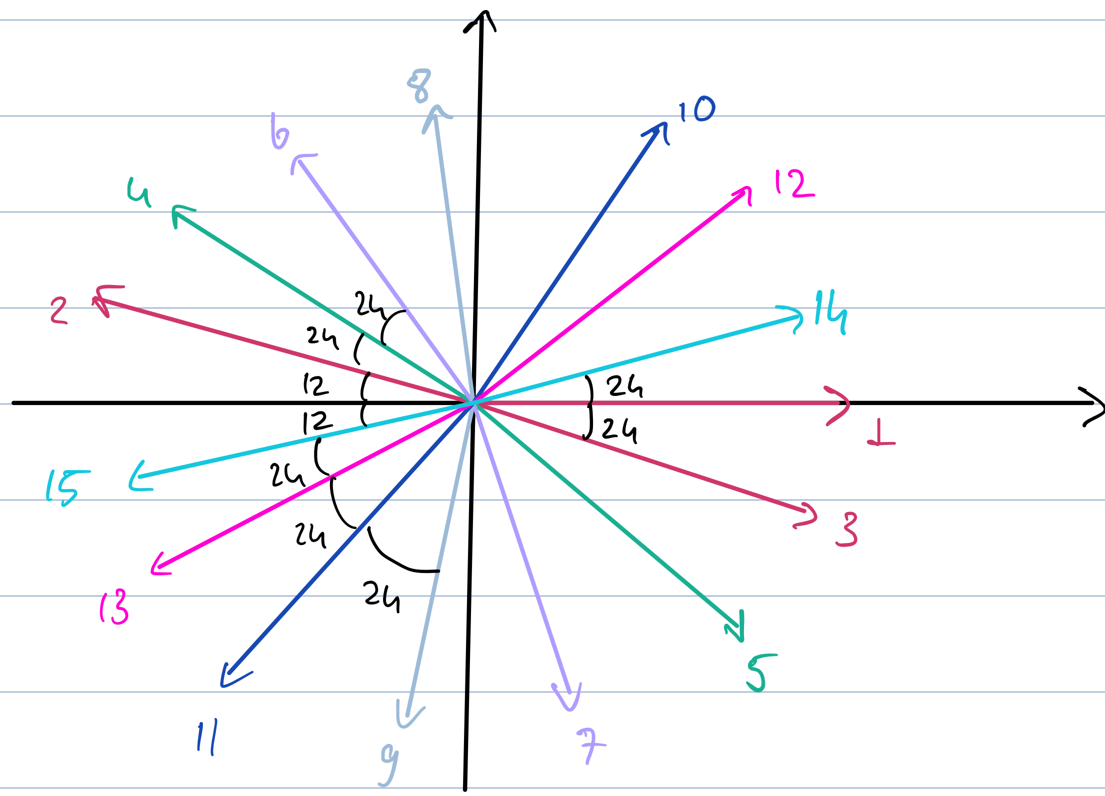
Q2

Number of slots : 15 (Q)

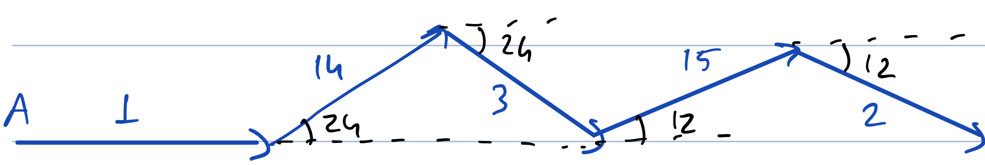
# of poles : 14 $\Rightarrow$ $p = 7$, $m = 3$

$q = \frac{Q}{p \cdot m} = \frac{5}{7}$ $\qquad$ $\alpha_u = 360 \cdot \frac{p}{Q} = 168^\circ$

Slot number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Elec. angle	0	168	-24	144	-48	120	-72	96	-96	72	-120	48	-144	24	-168



U_{f_A} :



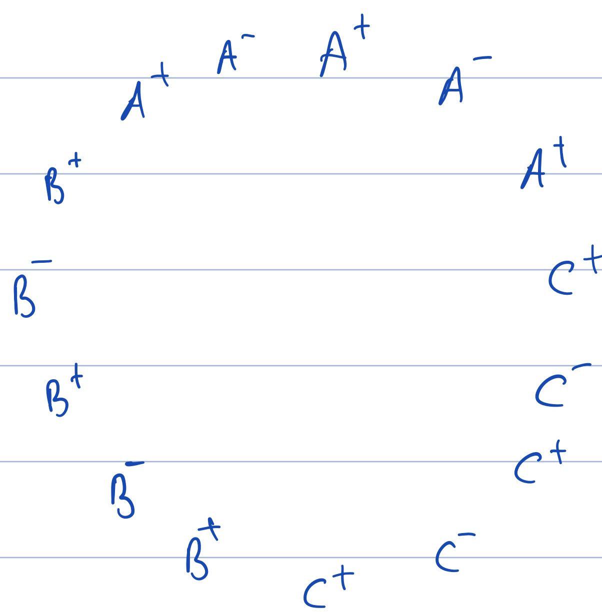
$$k_w = \frac{A \cdot (1 + 2 \cdot \sin(66) + 2\sin(78))}{5A} = 0,956$$

for 3^{rd} harmonic

$$k_w^3 = \frac{A.(1+2.\sin(18)+2.\sin(54))}{5.A} = 0{,}647$$

for 5^{th} harmonic

$$k_w^5 = \frac{A.(1+2.\cos(120)+2.\cos(60))}{5A} = 0{,}2$$



Q3

- Electric Loading : $\bar{A} = \frac{N_{turn}.I.Q}{\pi D_i}$, Q : # of slots

$I = J.A_{cu}$, A_{cu} = copper area

$A_{cu} = k_{wc}.A_{slot} = 0.6.\frac{\pi.(R_{so}^2 - R_{si}^2)}{15}.0{,}4$ mm^2 per slot

R_{so} : outer radius of stator $\quad R_{so} = 50$ mm

$R_{ro} = 0{,}62 \cdot 50 = 31 \text{ mm}$ (given)

$R_{si} = R_{ro} + l_m + g = 31 + 4 + 1 = 36 \text{ mm}$

* to keep the peak flux density close to 1,4 T, we approximately use 40% of the total area for copper winding.

$A_{cu} = 60{,}6 \text{ mm}^2$ per slot , $J = 5 \text{ A}_{rms}/\text{mm}^2$

$\Rightarrow I = 5 \cdot 60{,}6 = 303 \text{ A}_{rms}$

$$\bar{A} = \frac{15 \cdot 303 \text{ A}_{rms}}{\pi \cdot 2 \cdot 0{,}036} = 20\,093 \text{ A}_{rms}/\text{m}$$

- Carter's Coeff. $k_c = \dfrac{\tau_u}{\tau_u - K b_1}$, $K = \dfrac{b_1/\delta}{5 + b_1/\delta}$

tooth pitch $\tau_u = \dfrac{2\pi R_{si}}{N_s} = \dfrac{2\pi \cdot 36 \text{ mm}}{15} \cong 15 \text{ mm}$

slot opening $b_1 = 0{,}45 \cdot \tau_u \cong 6{,}75 \text{ mm}$

$\delta = g + \dfrac{l_m}{\mu_r} = 1 + \dfrac{4}{1{,}05} = 4{,}81 \text{ mm}$ effective air-gap

$K = \dfrac{6{,}75/4{,}8}{5 + 6{,}75/4{,}8} = 0{,}22$, $k_c = \dfrac{15}{15 - 0{,}22 \cdot 6{,}75} \cong 1{,}11$

effective air gap clearance : $\delta_e = k_c \cdot g = 1{,}11 \cdot 1 = 1{,}11 \text{ mm}$

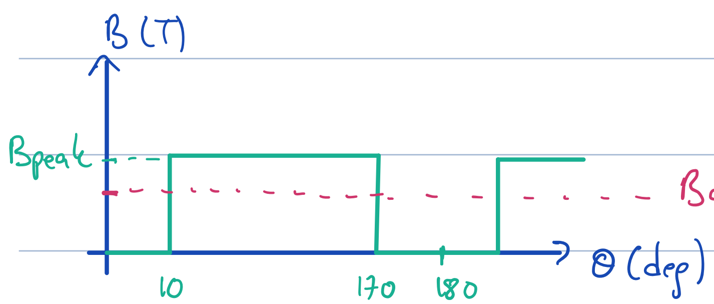
- $L_{eff} = L_s + 2g = 100 + 2 = 102 \text{ mm}$

- Air-gap peak flux density:

$$B_{peak} \approx \frac{B_r}{1 + \mu_r \cdot \frac{g}{d_M} \cdot k_c} = \frac{1{,}3}{1 + 1{,}05 \cdot \frac{1}{4} \cdot 1{,}11} \cong 1 \text{ T}$$

- Magnetic Loading:

$$\bar{B} = B_{peak} \cdot \alpha_{mp} \quad , \quad \alpha_{mp}: \text{magnet - pole ratio } (160/180)$$



$B_{avg} = \bar{B} = 1 \cdot 0{,}89 = 0{,}89 \text{ T}$

$$C = \frac{\pi^2}{\sqrt{2}} \cdot k_w \cdot \bar{A} \cdot \bar{B} = \frac{\pi^2}{\sqrt{2}} \cdot 0{,}956 \cdot 20\,093 \cdot 0{,}89 \cong 119\,31 \text{ J/m}^3$$

$\sigma = \bar{A} \cdot \bar{B}$ → from Adelaide University paper, from lecture notes

or

$\sigma = \frac{1}{\sqrt{2}} \cdot k_w \cdot \bar{A} \cdot B_{peak}$ → ai suggestion.

$$\sigma = \frac{1}{\sqrt{2}} \cdot 0{,}956 \cdot 1 \cdot 20\,093 = 13\,582 \text{ N/m}^2$$

$$T = 2\pi \cdot R_{si}^2 \cdot L_{eff} \cdot \sigma = 2\pi \cdot 0{,}036^2 \cdot 0{,}102 \cdot 13\,582 = 11{,}28 \text{ Nm}$$

- $\Phi = B_{av} \cdot \frac{2\pi R_{si} \cdot L_{eff}}{N_M} = 0{,}89 \cdot \frac{2\pi \cdot 0{,}036 \cdot 0{,}102}{14} = 1{,}466 \text{ mWb}$

$$N_{ph} = \frac{V_{rms}}{4{,}44 \cdot f \cdot k_w \cdot \Phi} = \frac{V_{rms} \cdot \sqrt{2}}{\omega \cdot k_w \cdot \Phi}$$

Assume Y connected machine. $V_{rms} \cdot \sqrt{2} = \frac{48}{\sqrt{3}} = 27{,}7\,V$

ω: $\omega_M = 1500 \cdot \frac{2\pi}{60} = 157$ rad/s

$\omega_e = p \cdot \omega_M = 7 \cdot 157 = 1099$ rad/s

$\Rightarrow N_{ph} \cong 18$